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A formula for the image intensity of phase objects in Zernike mode

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ABSTRACT

I present an analytical expression for the image intensity of a phase object visualized in Zernike phase contrast mode. The formula is valid for periodic and non-periodic weak and strong objects, and accounts for the effects of finite illumination. The expression provided is intended as a generalization of the standard reference formula given in the Born and Wolf [Principles of Optics, sixth ed., Pergamon Press, New York, 1980, p. 427] textbook as well as of the formalism employed to evaluate imaging doses in Zernike mode [M. Malac, M. Beleggia, R. Egerton, Y. Zhu, Ultramicroscopy 108 (2008) 126]. I illustrate the usefulness of the improved expression by means of three examples: a sinusoidal phase grating, a Gaussian object, and a phase step. The optimal Zernike phase angle yielding maximum image contrast is found to be object-dependent and not necessarily equal to $\pi/2$. Phase plate optimization criteria are derived and presented for two of the examples considered.

1. Introduction

With the advent of viable phase shifting elements providing Zernike-type contrast with electrons [3,4,1,2], it is essential to rely on a simple expression for computing the image intensity of phase objects, weak and strong (with electrons, strong phase objects are more common than with photons). In fact, image interpretation requires a reliable link between the physics of the sample, encoded in the electron-optical phase shift, and the experimental signal retrieved, i.e. the image intensity in Zernike mode.

The idea behind the well-known phase contrast method [5], which resulted in a Nobel prize for its inventor [6], is to introduce a controlled phase shift between unscattered and scattered portions of the image spectrum by means of a "phase plate". With photons, phase plates are thin glasses suitably prepared and placed in some focal plane of the imaging system; with electrons, they may be extremely thin (a few nm) films of electron-transparent materials such as carbon, with a tiny hole at the center, positioned in the diffraction plane of the microscope objective lens [7]. While in light optics, where phase objects are often very weak, the optimal shift is $\pi/2$, in electron optics this is not necessarily the case. Enhanced image contrast may be obtained by increasing the Zernike phase angle up to π when the phase object cannot be considered "weak". Tuning properly the phase plate according to the particular object under scrutiny requires the possibility of varying continuously the Zernike phase angle. This might be accomplished by utilizing electrostatic phase plates [1,8], where the phase shift is introduced by an external voltage.

While the standard expression describing image intensity in Zernike mode given by Born and Wolf [3] (hereafter referred to as "BW expression" or "BW formula") is a precious source of information, its validity is limited to weak phase objects. This fact may not be widely recognized because Born and Wolf appear to have neglected emphasizing the assumptions implicit in their treatment. The first part of this article outlines the limits of validity of the BW expression. In the following sections, I derive an improved expression and discuss its implications with three examples: an ideal phase grating, representative of an idealized atomic row in high resolution transmission electron microscopy, a Gaussian object with finite-size illumination, and a phase step, representative of a physical edge or abrupt thickness variation. Image simulations throughout the paper illustrate the qualitative and quantitative differences arising when employing the two formulas.

2. The Born–Wolf formula

Within the standard image-formation theory framework it is straightforward to derive an expression for the image intensity

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recorded in a microscope equipped with an ideal Zernike phase plate. Considering the electron wavefront associated to a generic phase object

$$\psi_0(x) = e^{i\varphi(x)} = 1 + [e^{i\varphi(x)} - 1], \quad (1)$$

and its Fourier Transform (object spectrum)

$$\tilde{\psi}_0(k) = \delta(k) + \mathscr{F}[e^{i\varphi(x)} - 1], \quad (2)$$

we can describe the effect of an ideal Zernike phase plate structured as an infinite transparent (e.g. non-absorbing) thin film with a central hole of negligible radius by leaving the $\delta(k)$ unchanged (axial unscattered electrons traveling through the hole), and multiplying the remaining part of the object spectrum by a phase factor $\exp(i\phi_Z)$. In other words, we introduce a phase plate transmission function as follows:

$$\begin{aligned} T_Z(k) &= 1 \quad \text{for } k = 0, \\ &= e^{i\phi_Z} \quad \text{for } |k| > 0. \end{aligned} \quad (3)$$

The Zernike angle ϕ_Z is proportional to the phase plate physical thickness t according to the standard expression [9]

$$\phi_Z = C_E V_0 t, \quad (4)$$

where C_E is the electrostatic interaction constant [9] and V_0 is the mean inner potential of the phase plate material (e.g. carbon).

We then obtain the image spectrum

$$\tilde{\psi}_i(k) = \delta(k) + e^{i\phi_Z}\mathscr{F}[e^{i\varphi(x)} - 1] \quad (5)$$

and, finally, the image intensity

$$I(x, \phi_Z) = |\psi_i(x)|^2 = |1 + e^{i\phi_Z}[e^{i\varphi(x)} - 1]|^2 \quad (6)$$

that can be simplified into a form equivalent to the BW [3]

$$I(x, \phi_Z) = 3 - 2\cos\phi_Z - 2\cos\varphi(x) + 2\cos[\phi_Z + \varphi(x)]. \quad (7)$$

While, apparently, the weak phase object approximation is not involved explicitly, it turns out that Eq. (7) is valid only for weak phase objects. In fact, the seemingly innocent mathematical "trick" of writing the initial phase object as in Eq. (1), that is to add and subtract 1, becomes troublesome when we apply the phase plate transmission function (3) because the coefficient multiplying the Dirac delta part of the spectrum is always equal to one. This is not necessarily the case, as it will be demonstrated shortly. The coefficient of the delta is indicative of the fraction of axial electrons passing through the hole unscattered. Only in the weak phase object approximation, that from this perspective is equivalent to weak/negligible high-angle scattering, we can safely assume that only a very small (to first order) fraction of electrons are not axial, so that the coefficient of the delta can be taken equal to one. Consequently, the intensity computed in this way suffers from an internal inconsistency, resulting in incorrect simulated intensity profiles. Furthermore, two additional troublesome implications are tacitly present in the BW formula:

- Charge conservation is violated.
- Constant phases become observable.

Normalization of image intensity over a periodicity (or over the beam size), i.e. the constraint

$$\langle I \rangle = \frac{1}{2L}\int_{-L}^{L} I(x)\,dx = 1, \quad (8)$$

is linked to charge conservation. In Eq. (8), $2L$ is the periodicity (or the beam size). Contrast features, such as dark and bright areas in the image, are created by re-distributing a given fixed number of illuminating electrons per unit length (or area, in the more general two-dimensional case). Hence, especially if we are interested in evaluating imaging doses quantitatively, e.g. the number of electrons required to develop a visible peak capable to overcome noise, we must avoid violating the normalization condition. Instead, the average BW-intensity

$$\langle I \rangle = 1 + 2(1 - \cos\phi_Z)[1 - \langle\cos\varphi\rangle] - 2\sin\phi_Z\langle\sin\varphi\rangle \neq 1, \quad (9)$$

depends on the Zernike phase angle ϕ_Z as well as on the object itself (more specifically, on the averaged sine and cosine of the phase).

Constant phases are, by definition, unobservable. Two phase objects that differs only by a constant are indistinguishable when imaged by any possible technique. This can be ascertained by considering the linearity of image formation: if $\psi_0(x) = \exp[i\varphi(x) + \varphi_0]$, where φ_0 is a constant phase factor, is the object wave, for any given transfer function $T(x)$ the image intensity is

$$I = |\psi_0(x) \otimes T(x)|^2 = |e^{i\varphi_0}e^{i\varphi(x)} \otimes T(x)|^2 = |e^{i\varphi(x)} \otimes T(x)|^2, \quad (10)$$

because $|\exp(i\varphi_0)|^2 = 1$. In other words, an additional constraint on the image intensity is that $I[\varphi(x) + \varphi_0] = I[\varphi(x)]$. It can be verified by direct evaluation

$$\begin{aligned} I[\varphi(x) + \varphi_0] = 3 &- 2\cos\phi_Z - 2\cos[\varphi(x) + \varphi_0] \\ &+ 2\cos[\phi_Z + \varphi(x) + \varphi_0] \neq I[\varphi(x)], \end{aligned} \quad (11)$$

that the BW expression does not satisfy such requirement.

3. The improved formula

To overcome the problems in the BW formula when applied to strong phase objects, we can revise and correct the "trick" [Eq. (1)] that was employed to generate Eq. (7). More specifically, rather than adding and subtracting 1, we add and subtract the correct weight of the delta function representing unscattered electrons, so that the internal inconsistency is removed. To illustrate conveniently the proposed computational strategy, we carry out the analysis in one dimension.

A generic periodic phase object $\exp[i\varphi(x)]$ with periodicity $2L$ can be expanded as a Fourier series

$$e^{i\varphi(x)} = \sum_{n=-\infty}^{+\infty} \gamma_n \exp\left(\frac{i\pi n x}{L}\right), \quad (12)$$

with a Fourier Transform

$$\mathscr{F}[e^{i\varphi(x)}] = \sum_{n=-\infty}^{+\infty} \gamma_n \delta(k - nk_0), \quad (13)$$

where $k_0 = 1/(2L)$. The coefficient of the $\delta(k)$ associated to unscattered electrons is, therefore, γ_0. Hence, we can simply add and subtract γ_0 rather than 1, and follow the same steps leading to the BW expression reported earlier. That is

$$\psi_0(x) = \gamma_0 + [e^{i\varphi(x)} - \gamma_0], \quad (14)$$

its Fourier Transform

$$\tilde{\psi}_0(k) = \gamma_0\delta(k) + \mathscr{F}[e^{i\varphi(x)} - \gamma_0], \quad (15)$$

the effect of the Zernike phase plate

$$\tilde{\psi}_i(k) = \gamma_0\delta(k) + e^{i\phi_Z}\mathscr{F}[e^{i\varphi(x)} - \gamma_0], \quad (16)$$

the image intensity

$$I(x, \phi_Z, \gamma_0) = |\gamma_0 + e^{i\phi_Z}[e^{i\varphi(x)} - \gamma_0]|^2, \quad (17)$$

which simplifies to

$$\begin{aligned} I(x, \phi_Z, \gamma_0) = 1 &+ 2\gamma^2 - 2\gamma^2\cos\phi_Z - 2\gamma\cos[\phi_\gamma + \varphi(x)] \\ &+ 2\gamma\cos[\phi_\gamma + \phi_Z + \varphi(x)], \end{aligned} \quad (18)$$

where γ_0, a complex number in general, has been written in its exponential form $\gamma_0 = \gamma\exp(-i\phi_\gamma)$. Note that when $\gamma_0 = 1$ we retrieve the BW expression (7). In general, γ_0 can be taken equal to one only for weak phase objects, because by its definition

$$\gamma_0 = \frac{1}{2L}\int_{-L}^{L} e^{i\varphi(x)}\,dx \sim \frac{1}{2L}\int_{-L}^{L}[1 + i\varphi(x) - \varphi^2(x) + \cdots]\,dx$$
$$= 1 + \frac{i}{2L}\int_{-L}^{L}\varphi(x)\,dx - \frac{1}{2L}\int_{-L}^{L}\varphi^2(x)\,dx + \cdots, \qquad (19)$$

the first non-vanishing contribution is at first order, whenever the phase profile does not average to zero over the periodicity, or second-order in all other situations. Hence, unless the phase object is weak in the sense just outlined, γ_0 differ from unity and the BW expression ceases to be valid. It can be ascertained that the improved expression (18) also satisfy charge conservation and the additional physical requirement associated to the inobservability of constant phases.

4. The phase grating

In order to expound the qualitative and quantitative differences between the BW formula and the expression just derived, we consider a simple phase grating as our test phase object:

$$\varphi(x) = \Phi \sin\left(\frac{\pi x}{L}\right), \qquad (20)$$

where Φ is the amplitude of the phase oscillations (phase oscillates from $-\Phi$ to $+\Phi$), and the periodicity is again $2L$. For this object, the coefficient γ_0 turns out to be a Bessel function

$$\gamma_0 = \frac{1}{2L}\int_{-L}^{L}\exp\left[i\Phi\sin\left(\frac{\pi x}{L}\right)\right]dx = J_0(\Phi). \qquad (21)$$

The image intensity can be evaluated directly from Eq. (18) with $\gamma = J_0(\Phi)$ and $\phi_\gamma = 0$ because γ_0 is real. A comparison between the BW formula (7) and the more general expression (18) is shown in Fig. 1 for several values of the phase grating angle $\Phi = \pi/10$ (a), $\pi/4$ (b), $\pi/2$ (c), π (d), with a fixed Zernike angle $\phi_Z = \pi/2$. As expected, the agreement is reasonable only whenever Φ is small, as in (a), i.e. in the weak phase object approximation. Note in particular the dramatic differences visible in (d), where contrast reversal occurs in addition to an overwhelming variation in peak intensities.

As mentioned in the introduction, while in general Zernike phase contrast is commonly associated to a quarter-wave shift of $\pi/2$ (the value used for the simulations in Fig. 1), this choice is not always optimal in terms of final image contrast. While it is true that the highest contrast is found for $\phi_Z = \pi/2$ when the phase grating is weak, in general a different Zernike angle is needed to have the highest possible visibility. For instance, when $\Phi = \pi/4$, the optimal Zernike angle turns out to be about 1.767 rad (while $\pi/2 = 1.571$ rad). This reflects a rather general rule for phase gratings that we can now demonstrate in light of the new expression. Limiting the parameter range as $\Phi \in (0, \Phi_0)$ where $\Phi_0 = 2.405$ rad is the first zero of the Bessel function $J_0(\Phi)$ [the image intensity is flat for $\Phi = \Phi_0$ independently on the choice of ϕ_Z because $\gamma = 0$ in Eq. (18)], and $\phi_Z \in (\pi/2, \pi)$, we observe that the peak intensity is always found at position $x = -L/2$, where the phase is equal to $-\Phi$. Since there is always some position along the intensity profile with values close to zero, to maximize contrast we only need to maximize the peak intensity. Hence, by taking the derivative of the peak value

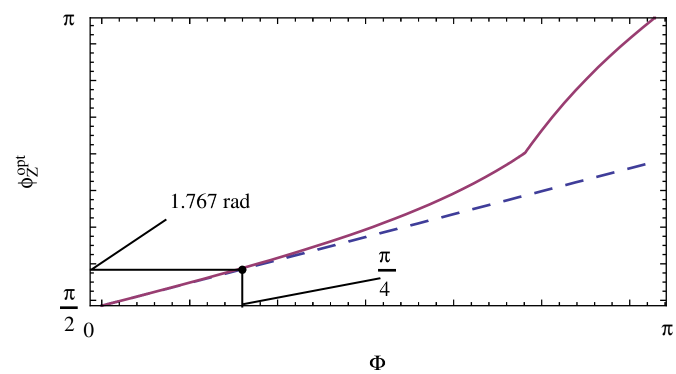

Fig. 2. Phase plate optimization criterion for phase gratings. The optimal Zernike phase angle of 1.767 rad for a grating with $\Phi = \pi/4$ is emphasized. The dashed line represents the linearized criterion valid only for small Φ.

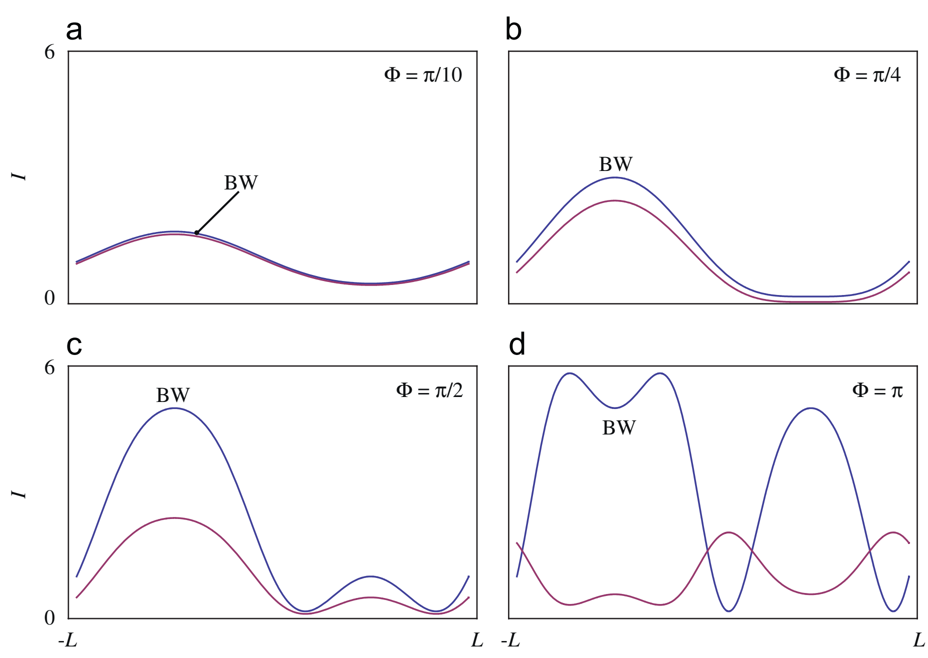

Fig. 1. Image intensity profiles for phase grating with $\Phi = \pi/10$ (a), $\pi/4$ (b), $\pi/2$ (c), π (d), according to the BW formula (Eq. (7), profiles labeled "BW") and to the improved expression (Eq. (18), unlabeled profiles). The Zernike phase angle is kept fixed at $\pi/2$.

$$I_{\max} = I(-L/2, \Phi, \phi_Z) = 1 + 2\gamma[(1 - \cos\phi_Z)(\gamma - \cos\Phi) + \sin\Phi\sin\phi_Z]$$

$$\frac{\partial}{\partial\phi_Z} I_{\max} = 2\gamma[(\gamma - \cos\Phi)\sin\phi_Z + \sin\Phi\cos\phi_Z], \tag{22}$$

and finding its root, we obtain a definitory relationship for the optimal Zernike angle

$$\phi_Z^{\text{opt}} = \frac{\pi}{2} + \operatorname{arccot}\left(\frac{\sin\Phi}{\gamma - \cos\Phi}\right), \tag{23}$$

that is valid for phase gratings where $\Phi < \Phi_0$. A plot of the optimal Zernike angle is reported in Fig. 2, accompanied (dashed line) by the linear expansion $\phi_Z^{\text{opt}} = \pi/2 + \Phi/4$ valid for small Φ and justifying the result obtained earlier that when $\Phi = \pi/4$ the highest contrast is obtained for $\phi_Z \sim \pi/2 + \pi/16 = 1.767$ rad.

5. Non-periodic objects

When the phase shift is bounded within some spatial interval, as when the object is bell-shaped or, more in general, when it reflects a finite-size isolated object, the factor γ_0 can be expressed as

$$\gamma_0 = 1 + \frac{1}{2L}\int_{-a}^{a} [e^{i\varphi(x)} - 1]\,dx. \tag{24}$$

That is, one plus a complex number that depends on the beam (aperture) size $2L$ and on the extension of the object $2a$. Whenever the beam/aperture size is much larger than the object size ($L \gg a$), we have $\gamma_0 \to 1$. As demonstrated above, when γ_0 is close to one, the BW formula is applicable. From this perspective, non-periodic objects appear "simpler" in some sense, and the improved formula less useful. However, particular care is needed when dealing with the effects of the illumination size, as related to charge conservation. We have already seen how the BW formula violates the requirement of the averaged intensity being equal to one. So, it may seem unclear why now calculations suggest that whenever $L \gg a$, the resulting image intensity that does not average to one is indeed correct.

5.1. The Gaussian object

To illustrate this issue, we consider a simple Gaussian object $\varphi(x) = \Phi\exp(-x^2/x_0^2)$ (where its extension $2a$ can be roughly taken as $2a = 10x_0$), set $\Phi = \pi/3$, and evaluate the image intensity in Zernike mode with $\phi_Z = \pi/2$ varying the illumination size $2L$ from $2a$ ($L = a$) to $4a$ ($L > a$) to $10a$ ($L \gg a$). The simulations with the BW and improved formulas are shown in Fig. 3.

We observe that, apparently, the improved formula "converges" to the BW formula, the main difference being the background intensity level: while the BW, that does not depend on L, shows a background level of 1, the improved formula shows an L-dependent saturation level, with a value closer and closer to one as we increase the illumination size. Since charge conservation is a physical requirement, the electrons contributing to the central peak formation must necessarily come from somewhere else in the visible portion of the image. This is more clearly visualized by considering the shaded areas in the simulations: the area under the peak (lighter gray) must always be equal to the area between 1 and the saturation level (darker gray). Consequently, as we increase L the central peak develops at the expense of a smaller and smaller fraction of background intensity over a larger and larger interval. We have so verified that, for non-periodic objects, the BW formula is a valid approximation whenever the illumination size is much larger than the object size. Since this scenario is not universal, as some interesting experimental setups are conceivable where the spot size is just large enough to cover the object, the usefulness of the improved formula extends to describe such situations correctly.

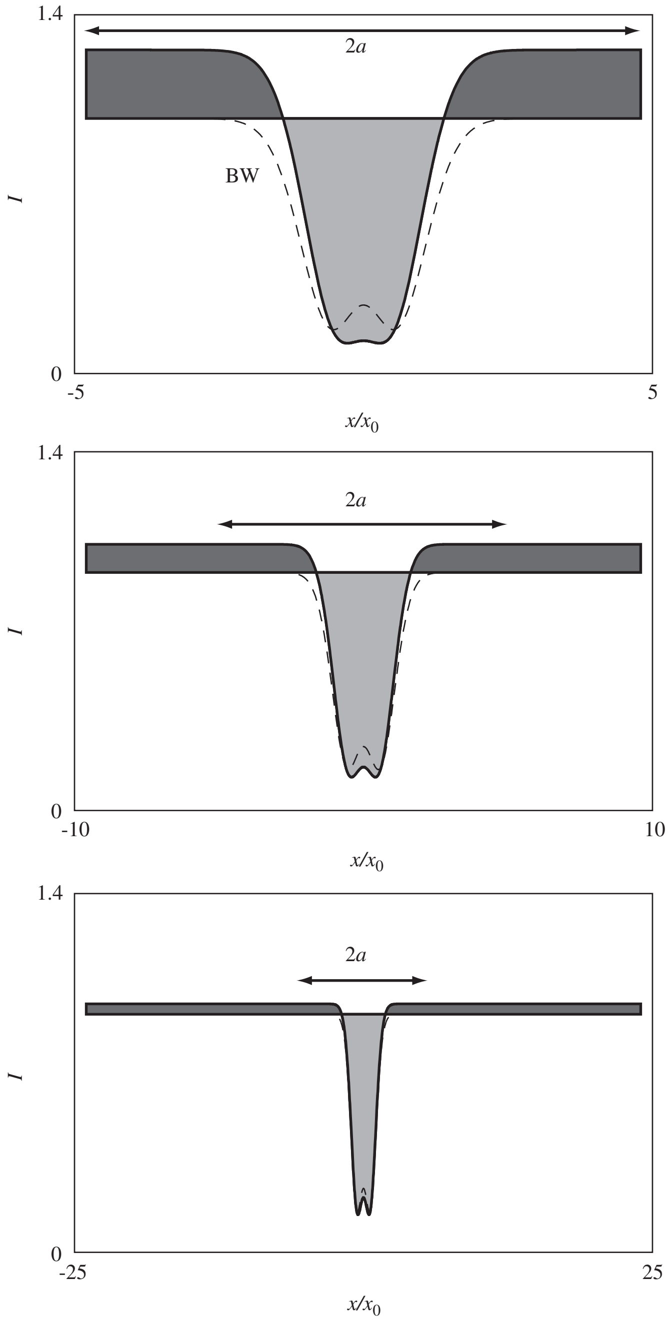

Fig. 3. Effect of finite illumination on the image intensity of a Gaussian object $\Phi\exp(-x^2/x_0^2)$ with $\Phi = \pi/3$. From top to bottom: $2L = 10x_0 = 2a$, $2L = 20x_0 = 4a$, $2L = 50x_0 = 10a$. Dashed line is the intensity profile calculated according to the BW formula, which does not depend on the illumination size.

5.2. The phase step

Phase steps are often encountered in experiments, either at the specimen edge, or at some interface, or when the sample has some non-trivial thickness profile. Since phase steps are, in good approximation, phase objects, they are invisible unless a phase contrast method is employed. In the work of Huang et al. [1], Zernike mode was employed to enhance the phase contrast at a $SiON/SiO_2$ interface. The cited authors found a measurable contrast enhancement when the phase plate (with $\phi_Z \sim \pi/2$) was inserted and played its part in image formation. In this section, we illustrate how the image intensity in Zernike mode varies as the phase step is increased, and the dependence of the peak intensity value on the choice of the Zernike plate. We establish the criterion for phase plate optimization, which turns out to be rather different than the one for phase grating [Eq. (23)], and conclude with the solution of the inverse problem formulated as follows: if we measure a peak intensity value I, can we relate its value to the phase step Φ?

We define the phase step as an object, where

$$\begin{aligned} \varphi(x) &= \Phi \quad \text{for } |x|<a, \\ &= 0 \quad \text{for } |x|>a. \end{aligned} \tag{25}$$

as illustrated in Fig. 4(a). We assume the illumination size to be much larger than a, so that $\gamma_0 \sim 1$ from Eq. (24), and we can freely utilize either the BW or the improved formula. The image intensity is straightforward: a constant background of 1, and a central peak (of width $2a$) with intensity value

$$I(\Phi, \phi_Z) = 3 - 2\cos\phi_Z - 2\cos\Phi + 2\cos(\phi_z + \Phi). \tag{26}$$

An example of intensity profile is shown in Fig. 4(a) with $\Phi = \pi/4$ and $\phi_Z = 3\pi/2$.

In order to maximize the peak intensity, we choose to work around the "negative" phase contrast condition (ϕ_Z around $3\pi/2$), where the object is visualized as a bright peak, but similar considerations apply for the "positive" phase contrast condition where the peak is dark (ϕ_Z around $\pi/2$). A two-dimensional representation of the peak intensity as a function of Φ and ϕ_Z is displayed in Fig. 4(b). Superimposed is a dashed line where the peak intensity reaches its maximum value, corresponding to the optimal choice of ϕ_Z given a specific Φ, and the dot-dashed line representing the profile at $\phi_Z = 3\pi/2$. By searching for the maximum of Eq. (26), we find the equation of such line, i.e. the phase plate optimization criterion valid for phase steps:

$$\phi_Z^{\text{opt}} = \frac{3}{2}\pi - \frac{\Phi}{2}. \tag{27}$$

When the phase plate is tuned according to Eq. (27), the image intensity simplifies to

$$I(\Phi) = \left(1 + 2\sin\frac{\Phi}{2}\right)^2 \tag{28}$$

that can reach the extreme value of 9 if the phase step is exactly one π (for which the optimum ϕ_Z is as well π). The profile of the optimized intensity is shown in Fig. 4(c), compared with the profile keeping ϕ_Z fixed at the value $3\pi/2$ [dot-dashed line in (b)].

Since Eq. (28) is invertible in analytical form as

$$\Phi = 2\arcsin\left(\frac{\sqrt{I} - 1}{2}\right), \tag{29}$$

we are offered the opportunity of solving the inverse problem directly, and upgrade Zernike mode from a "phase contrast" to a "phase retrieval" method. In fact, if in the experiment we observe a certain peak intensity I, from Eq. (29) we can quantitatively and univocally (for $0<\Phi<\pi$) relate such value with the phase step. The inverse function equation (29) is shown in Fig. 4(d). Clearly, since the inverse relationship was derived in the optimal condition $\phi_Z = \phi_Z^{\text{opt}}$, we need the ability to tune continuously the phase plate in order to maximize experimentally the peak intensity before actually extracting the measurement. This cannot be accomplished with thin-film type phase plates, and requires instead tunability such as with electrostatic plates. Nevertheless, even without tunable phase plates, by keeping the Zernike angle at a constant value, we can still extract information on the phase step value. However, the inversion is not as straightforward, and, more importantly, not univocal. This is briefly shown in Fig. 4(c), where an illustrative intensity value of 5.5 may be related with either $\Phi = 1.87$ rad or $\Phi = 2.84$ rad.

6. Conclusions

Zernike phase contrast with electrons may well be the next breakthrough in transmission electron microscopy. Similarly to what occurred with photons, a whole new range of materials and systems, especially at the interface between soft and hard matter (single molecules, functionalized nanostructures, radiation sensitive materials), may soon be studied routinely at resolutions near the current limits. However, solid image interpretation is required to extract reliable physical information from the experiments. In this paper, I have presented an analytical formula describing the Zernike image intensity distribution associated to generic phase objects. Compared to the standard reference expression of Born and Wolf [3], which

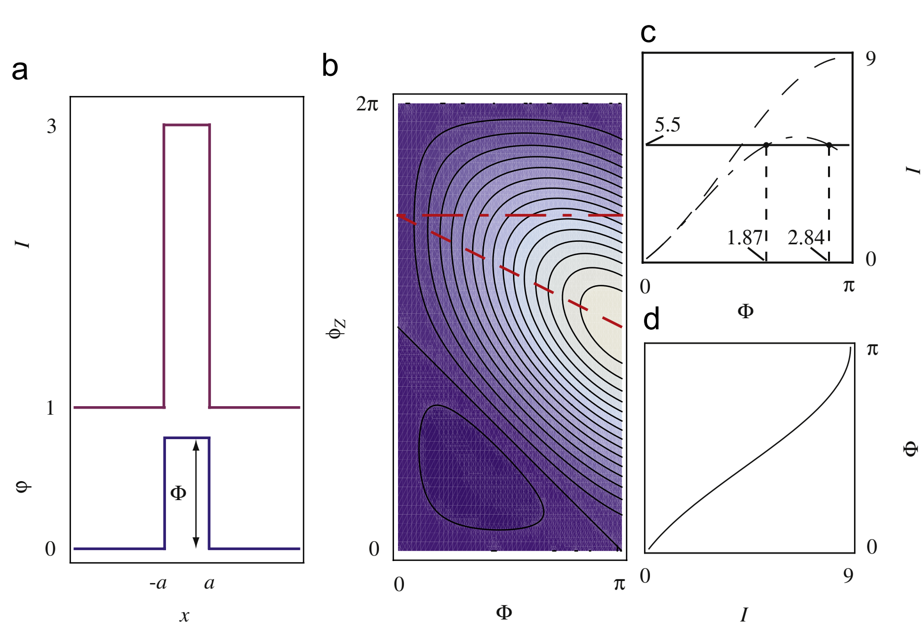

Fig. 4. (a) Phase step and corresponding image intensity when $\Phi = \pi/4$ and $\phi_Z = 3\pi/2$; (b) peak intensity value as a function of Φ and ϕ_Z; (c) intensity profiles for the optimized phase plate $\phi_Z = \phi_Z^{\text{opt}} = 3\pi/2 - \Phi/2$ (dashed line) and for $\phi_Z = 3\pi/2$ (dot-dashed line); (d) plot of the inverse formula, Eq. (29).

fails when strong periodic phase objects are involved, the new formula accounts correctly for weak and strong phase objects, and satisfies the physical requirements of charge conservation and inobservability of constant phases. Examples illustrated explicitly the main features and usefulness of the improved expression. Criteria for phase plate optimization showed that the commonly accepted quarter-wave shift may not necessarily be the best choice. It is my hope that these theoretical considerations will support interpretation of actual experimental data and, eventually, stimulate further progress in the field of Zernike phase contrast with electrons.

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